

CIS 5200 – Final Code

Analysis of U.S. Used Car Market Trends Using Hadoop and Hive

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-- STEP 1: Run on LOCAL machine CLI

cd downloads

scp archive.zip teremio@132.226.148.236:/tmp

-- STEP 2: SSH into server

ssh teremio@132.226.148.236

-- Confirm file is there

ls -lh /tmp

-- Unzip

unzip /tmp/archive.zip -d /tmp

-- Upload to HDFS

hdfs dfs -mkdir /tmp/car

hdfs dfs -put /tmp/used_cars_data.csv /tmp/car

hdfs dfs -ls /tmp/car

-- Confirm CSV is in HDFS then delete from server

rm /tmp/archive.zip

rm /tmp/used_cars_data.csv

OPEN ANOTHER WINDOW FOR THIS STEP

-- STEP 3: Connect to Hive and Create External Table--

ssh teremio@132.226.148.236

beeline

use teremio;

```
DROP TABLE IF EXISTS used_cars_raw;
```

```
CREATE EXTERNAL TABLE used_cars_raw (  
    vin STRING, back_legroom STRING, bed STRING, bed_height STRING,  
    bed_length STRING, body_type STRING, cabin STRING, city STRING,  
    city_fuel_economy STRING, combine_fuel_economy STRING, daysonmarket STRING,  
    dealer_zip STRING, description STRING, engine_cylinders STRING,  
    engine_displacement STRING, engine_type STRING, exterior_color STRING,  
    fleet STRING, frame_damaged STRING, franchise_dealer STRING,  
    franchise_make STRING, front_legroom STRING, fuel_tank_volume STRING,  
    fuel_type STRING, has_accidents STRING, height STRING,  
    highway_fuel_economy STRING, horsepower STRING, interior_color STRING,  
    isCab STRING, is_certified STRING, is_cpo STRING, is_new STRING,  
    is_oemcpo STRING, latitude STRING, length STRING, listed_date STRING,  
    listing_color STRING, listing_id STRING, longitude STRING,  
    main_picture_url STRING, major_options STRING, make_name STRING,  
    maximum_seating STRING, mileage STRING, model_name STRING,  
    owner_count STRING, power STRING, price STRING, salvage STRING,  
    savings_amount STRING, seller_rating STRING, sp_id STRING,  
    sp_name STRING, theft_title STRING, torque STRING, transmission STRING,  
    transmission_display STRING, trimId STRING, trim_name STRING,  
    vehicle_damage_category STRING, wheel_system STRING,  
    wheel_system_display STRING, wheelbase STRING, width STRING, year STRING  
)  
  
ROW FORMAT SERDE 'org.apache.hadoop.hive.serde2.OpenCSVSerde'  
  
WITH SERDEPROPERTIES (  
    "separatorChar" = ",",  
    "quoteChar" = "\"",
```

```

"escapeChar" = "\\"
)

STORED AS TEXTFILE

LOCATION '/tmp/car'

TBLPROPERTIES ("skip.header.line.count"="1");

-- View your table--

SELECT * FROM used_cars_raw LIMIT 5;
DESCRIBE FORMATTED used_cars_raw;

-- STEP 4: Create Clean Table--

DROP TABLE IF EXISTS used_cars_clean;

CREATE TABLE used_cars_clean AS

SELECT

    CAST(latitude AS DOUBLE)          AS latitude,

    CAST(longitude AS DOUBLE)         AS longitude,

    city,

    YEAR(TO_DATE(listed_date))        AS listing_year,

    MONTH(TO_DATE(listed_date))        AS listing_month,

    CAST(daysonmarket AS INT)          AS days_on_market,

    CASE

        WHEN CAST(daysonmarket AS INT) <= 20 THEN 'High Demand'

        WHEN CAST(daysonmarket AS INT) <= 45 THEN 'Medium Demand'

        ELSE 'Low Demand'

    END                                AS demand_category,

    make_name,

    fuel_type,

    CAST(price AS DOUBLE)              AS price,

    CAST(mileage AS DOUBLE)            AS mileage,

    CASE

```

```

        WHEN has_accidents = 'True' THEN true

        ELSE false

    END                AS has_accidents

FROM used_cars_raw

WHERE

    price IS NOT NULL

    AND CAST(price AS DOUBLE) > 500

    AND latitude IS NOT NULL

    AND longitude IS NOT NULL

    AND listed_date IS NOT NULL

    AND make_name IS NOT NULL

    AND daysonmarket IS NOT NULL

    AND fuel_type IS NOT NULL

    AND has_accidents IS NOT NULL

    AND mileage IS NOT NULL

    AND CAST(mileage AS DOUBLE) > 0

LIMIT 500000;


-- Verify clean table--

SELECT * FROM used_cars_clean LIMIT 5;
DESCRIBE FORMATTED used_cars_clean;


-- QUERY 1 : Demand by Fuel Type Over Time--

SELECT

    TRIM(fuel_type)                AS fuel_type,

    listing_year,

    COUNT(*)                       AS total_listings,

    ROUND(AVG(days_on_market), 1) AS avg_days_on_market

FROM used_cars_clean

```

WHERE

listing_year >= 2016

AND fuel_type IS NOT NULL

AND TRIM(fuel_type) <> ''

AND days_on_market IS NOT NULL

GROUP BY

TRIM(fuel_type),

listing_year

HAVING

COUNT(*) >= 50

ORDER BY

listing_year,

total_listings DESC

LIMIT 20;

-- QUERY 2 : Accident History Impact on Price and Demand--

SELECT

has_accidents,

COUNT(*) AS total_listings,

ROUND(AVG(price), 2) AS avg_price,

ROUND(AVG(days_on_market), 1) AS avg_days_on_market

FROM used_cars_clean

GROUP BY

has_accidents

ORDER BY

avg_price DESC;

-- QUERY 3 : Top 10 Used Car Brands in the US--

SELECT

make_name,

```

COUNT(*)          AS total_listings,

ROUND(AVG(days_on_market), 1) AS avg_days_on_market

FROM used_cars_clean

GROUP BY

    make_name

HAVING

    COUNT(*) >= 500

ORDER BY

    avg_days_on_market ASC,

    total_listings DESC

LIMIT 10;

```

-- QUERY 4 : Distribution of Car Sales by State (Spatial)--

```

SELECT

    city,

    ROUND(latitude, 4)          AS latitude,

    ROUND(longitude, 4)        AS longitude,

    COUNT(*)                   AS total_listings,

    ROUND(AVG(price), 2)       AS avg_price,

    ROUND(AVG(days_on_market), 1) AS avg_days_on_market

FROM used_cars_clean

GROUP BY

    city,

    ROUND(latitude, 4),

    ROUND(longitude, 4)

ORDER BY

    total_listings DESC

LIMIT 20;

```

-- QUERY 5 : Tempo-Spatial Seasonal Trends by Month and Location--

SELECT

city,

ROUND(latitude, 4) AS latitude,

ROUND(longitude, 4) AS longitude,

CONCAT(listing_year, '-',

LPAD(CAST(listing_month AS STRING), 2, '0'),

'-01') AS listing_date,

COUNT(*) AS total_listings,

ROUND(AVG(days_on_market), 1) AS avg_days_on_market,

demand_category

FROM used_cars_clean

GROUP BY

city,

ROUND(latitude, 4),

ROUND(longitude, 4),

CONCAT(listing_year, '-',

LPAD(CAST(listing_month AS STRING), 2, '0'),

'-01'),

demand_category

ORDER BY

listing_date,

total_listings DESC

LIMIT 20;

ORDER BY

fuel_type,

avg_mileage ASC

LIMIT 20;

-- STEP 6: Export All Queries to CSV--

-- EXPORT QUERY 1: Demand by Fuel Type Over Time

INSERT OVERWRITE DIRECTORY '/tmp/teremio/fueltype_demand'

ROW FORMAT DELIMITED FIELDS TERMINATED BY ','

SELECT

TRIM(fuel_type) AS fuel_type,

listing_year,

COUNT(*) AS total_listings,

ROUND(AVG(days_on_market), 1) AS avg_days_on_market

FROM used_cars_clean

WHERE

listing_year >= 2016

AND fuel_type IS NOT NULL

AND TRIM(fuel_type) <> ''

AND days_on_market IS NOT NULL

GROUP BY

TRIM(fuel_type),

listing_year

HAVING

COUNT(*) >= 50

ORDER BY

listing_year,

total_listings DESC;

-- EXPORT QUERY 2: Accident History Impact on Price and Demand

INSERT OVERWRITE DIRECTORY '/tmp/teremio/accident_impact'

ROW FORMAT DELIMITED FIELDS TERMINATED BY ','


```

SELECT

    has_accidents,

    COUNT(*)          AS total_listings,

    ROUND(AVG(price), 2)    AS avg_price,

    ROUND(AVG(days_on_market), 1) AS avg_days_on_market

FROM used_cars_clean

GROUP BY

    has_accidents

ORDER BY

    avg_price DESC;

```

-- EXPORT QUERY 3: Top 10 Used Car Brands in the US

```

INSERT OVERWRITE DIRECTORY '/tmp/teremio/top_brands'

```

```

ROW FORMAT DELIMITED FIELDS TERMINATED BY ','

```

```

SELECT

    make_name,

    COUNT(*)          AS total_listings,

    ROUND(AVG(days_on_market), 1) AS avg_days_on_market

FROM used_cars_clean

GROUP BY

    make_name

HAVING

    COUNT(*) >= 500

ORDER BY

    avg_days_on_market ASC,

    total_listings DESC

LIMIT 10;

```

-- EXPORT QUERY 4: Distribution of Car Sales by State (Spatial)

```
INSERT OVERWRITE DIRECTORY '/tmp/teremio/sales_by_state'
```

```
ROW FORMAT DELIMITED FIELDS TERMINATED BY ','
```

```
SELECT
```

```
    city,
```

```
    ROUND(latitude, 4)      AS latitude,
```

```
    ROUND(longitude, 4)    AS longitude,
```

```
    COUNT(*)               AS total_listings,
```

```
    ROUND(AVG(price), 2)   AS avg_price,
```

```
    ROUND(AVG(days_on_market), 1) AS avg_days_on_market
```

```
FROM used_cars_clean
```

```
GROUP BY
```

```
    city,
```

```
    ROUND(latitude, 4),
```

```
    ROUND(longitude, 4)
```

```
ORDER BY
```

```
    total_listings DESC;
```

```
-- EXPORT QUERY 5: Tempo-Spatial Seasonal Trends by Month and Location
```

```
INSERT OVERWRITE DIRECTORY '/tmp/teremio/seasonal_trends'
```

```
ROW FORMAT DELIMITED FIELDS TERMINATED BY ','
```

```
SELECT
```

```
    city,
```

```
    ROUND(latitude, 4)      AS latitude,
```

```
    ROUND(longitude, 4)    AS longitude,
```

```
    CONCAT(listing_year, '-',
```

```
        LPAD(CAST(listing_month AS STRING), 2, '0'),
```

```
        '-01')              AS listing_date,
```

```
    COUNT(*)               AS total_listings,
```

```
    ROUND(AVG(days_on_market), 1) AS avg_days_on_market,
```

```

demand_category
FROM used_cars_clean
GROUP BY
    city,
    ROUND(latitude, 4),
    ROUND(longitude, 4),
    CONCAT(listing_year, '-',
        LPAD(CAST(listing_month AS STRING), 2, '0'),
        '-01'),
    demand_category
ORDER BY
    listing_date,
    total_listings DESC;

```

-- EXPORT QUERY 6: Average price by fuel type and mileage

INSERT OVERWRITE DIRECTORY '/tmp/teremio/fuel_mileage'

ROW FORMAT DELIMITED FIELDS TERMINATED BY ','

SELECT

```

    TRIM(fuel_type)          AS fuel_type,

CASE
    WHEN mileage < 20000 THEN '0-20k'
    WHEN mileage < 40000 THEN '20k-40k'
    WHEN mileage < 60000 THEN '40k-60k'
    WHEN mileage < 80000 THEN '60k-80k'
    WHEN mileage < 100000 THEN '80k-100k'
    ELSE '100k+'

END          AS mileage_bucket,

COUNT(*)   AS total_listings,

```

```

ROUND(AVG(price), 2)      AS avg_price,

ROUND(AVG(mileage), 0)    AS avg_mileage

FROM used_cars_clean

WHERE

    mileage IS NOT NULL

    AND mileage > 0

    AND fuel_type IS NOT NULL

    AND TRIM(fuel_type) <> ''

    AND TRIM(fuel_type) <> 'N/A'

GROUP BY

    TRIM(fuel_type),

CASE

    WHEN mileage < 20000 THEN '0-20k'

    WHEN mileage < 40000 THEN '20k-40k'

    WHEN mileage < 60000 THEN '40k-60k'

    WHEN mileage < 80000 THEN '60k-80k'

    WHEN mileage < 100000 THEN '80k-100k'

    ELSE '100k+'

END

ORDER BY

    fuel_type,

    avg_mileage ASC;

```

-- STEP 7: Pull files from HDFS to server home directory

```
hdfs dfs -get /tmp/teremio/fueltype_demand/000000_0 fueltype_demand.csv
```

```
hdfs dfs -get /tmp/teremio/accident_impact/000000_0 accident_impact.csv
```

```
hdfs dfs -get /tmp/teremio/top_brands/000000_0 top_brands.csv
```

```
hdfs dfs -get /tmp/teremio/sales_by_state/000000_0 sales_by_state.csv
```

```
hdfs dfs -get /tmp/teremio/seasonal_trends/000000_0 seasonal_trends.csv
```

```
hdfs dfs -get /tmp/teremio/fuel_mileage/000000_0 fuel_mileage.csv
```

```
ls -lh
```

-- STEP 8: SCP files to your local machine (run on LOCAL CLI)--

```
scp teremio@132.226.148.236:~/fueltype_demand.csv C:\Users\thelm\downloads\
```

```
scp teremio@132.226.148.236:~/accident_impact.csv C:\Users\thelm\downloads\
```

```
scp teremio@132.226.148.236:~/top_brands.csv C:\Users\thelm\downloads\
```

```
scp teremio@132.226.148.236:~/sales_by_state.csv C:\Users\thelm\downloads\
```

```
scp teremio@132.226.148.236:~/seasonal_trends.csv C:\Users\thelm\downloads\
```

```
scp teremio@132.226.148.236:~/fuel_mileage.csv C:\Users\thelm\downloads\
```